

Does Your MVP Actually Work?

Testing and Validation for Engineering Products

Entrepreneurship Cohort Program (E-Co-Op)

C-SHInE | KLE Technological University

SETTING THE STAGE

"Our MVP Works!"

What exactly does that mean?

? It switches ON?

? It performed the intended function once?

? It worked during the demonstration?

? It gave the expected output?

Is that enough evidence?

A working prototype is a claim. Testing provides the evidence.

What Does "It Works" Really Mean?

Five questions we need to answer

F

Functionality

Does it do what it is supposed to do?

P

Performance

How well does it do it?

R

Reliability

Does it continue to work consistently?

U

Usability

Can the intended user use it effectively?

A

User Acceptance

Does it satisfy the user's/customer's needs sufficiently for acceptance?

STEP ONE

Start With the Functions of Your MVP

What should your product actually DO?

Computer Science

- Identify a registered user from an image
- Generate a report from uploaded data

Electronics

- Measure temperature
- Detect the presence of an object

Mechanical

- Lift a specified load
- Dispense a specified quantity of material

Automation / Robotics

- Detect an obstacle and stop
- Pick and place an object

Testing starts by identifying the critical functions of your MVP.

Functionality Testing

Does it do what it is supposed to do?

Functionality testing checks whether the intended functions of the MVP actually work.

Function	Functionality Test
Identify a registered user	Present registered and unregistered users
Measure temperature	Expose sensor to known temperatures
Lift a load	Apply the intended load and operate mechanism
Detect an obstacle and stop	Place obstacle in robot's path

FUNCTIONALITY = Does the intended function happen?

REALITY CHECK

But "It Works" Is Not Enough

Consider an obstacle-detecting robot...

Function: Detect an obstacle and stop.

- It detects the obstacle.
- It stops.

But...

- At what distance does it detect?
- How accurately does it detect?
- How quickly does it respond?
- How far does it travel before stopping?
- Does lighting affect detection?

We now need PERFORMANCE TESTING.

Performance Testing

How well does it work?

Performance testing measures quantifiable characteristics of the MVP under defined operating conditions.

■ Computer Science

- Accuracy
- Response time
- Processing speed
- Error rate

■ Electronics

- Measurement accuracy
- Response time
- Power consumption
- Operating range

■ Mechanical

- Load capacity
- Speed
- Efficiency
- Dimensional accuracy

■ Automation / Robotics

- Positioning accuracy
- Cycle time
- Detection range
- Stopping distance

The right metric depends on YOUR product.

From Function to Performance

Domain	Function	Performance Metric
Computer Science	Identify a face	Recognition accuracy, response time
Electronics	Measure temperature	Measurement error, response time
Mechanical	Lift a load	Maximum load, lifting time
Automation	Detect obstacle and stop	Detection range, stopping distance

Functionality asks:

Does it happen?

Performance asks:

How well does it happen?

Measuring Something Is Not Enough

“We will measure detection accuracy.”

What accuracy is acceptable?

“We will measure response time.”

What response time is acceptable?

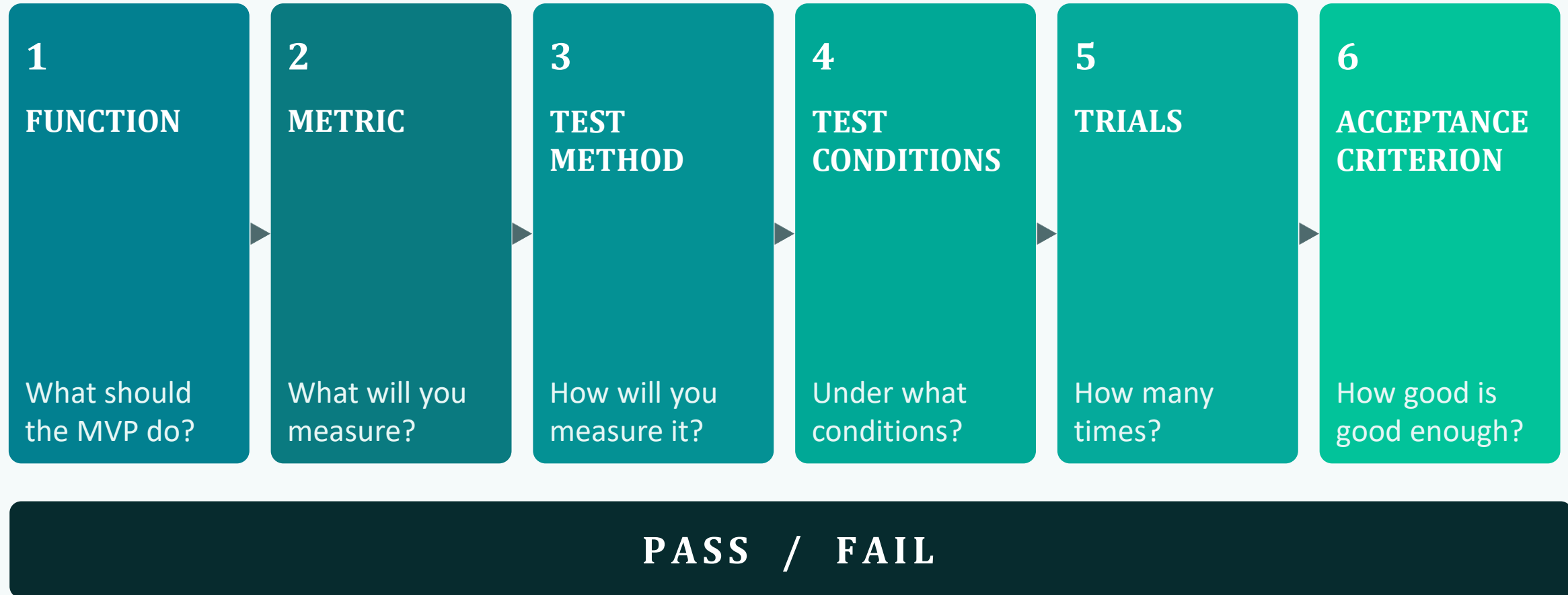
“We will test the maximum load.”

What load should the product handle?

Every test needs an ACCEPTANCE CRITERION.

Acceptance Criterion = The result that must be achieved for the test to PASS.

Anatomy of a Good Test



WORKED EXAMPLE

Example of a Complete Performance Test

Function	Detect an obstacle and stop the robot.
Metric	Detection success rate
Test Method	Place an obstacle at predefined distances and record successful detections.
Test Conditions	10 cm, 20 cm and 30 cm • Normal indoor lighting
Number of Trials	10 trials at each distance
Acceptance Criterion	≥ 95% successful detection

Result: PASS / FAIL

Note: Values shown are illustrative. Your acceptance criteria must be justified for your product.

Where Should Acceptance Criteria Come From?

Don't invent a number because it "looks reasonable."



Validated user requirements



Engineering/design requirements



Industry standards



Safety requirements



Competitor/product benchmarks



Engineering calculations



Manufacturer specifications



Experimental evidence

Every criterion must be traceable to a real source — not a guess.

Your question should be:

"Why is THIS level of performance acceptable?"

Prove That Your MVP Works

- Identify 3 to 5 critical functions of your MVP.
- For each function, determine: functionality test, performance metric(s), test method, test conditions, number of trials, and acceptance criterion.

Function	Functionality Test	Performance Metric	Test Method	Conditions	Trials	Acceptance Criterion

Don't just write "Check whether it works."

REFLECTION

Can You Defend Your Test?

Be prepared to answer:

- 1 What exactly are you measuring?
- 2 Why did you choose that metric?
- 3 Why did you choose those test conditions?
- 4 Why that number of trials?
- 5 Where did your acceptance criterion come from?

A number without justification is just a number.

Your MVP Passed. Are We Done?

Imagine:

- ✓ Your software correctly processed the data.
- ✓ Your electronic sensor measured accurately.
- ✓ Your mechanism lifted the required load.
- ✓ Your robot detected and avoided the obstacle.

Now repeat it...

100 times.

500 times.

1,000 times.

Will it still work?

Reliability Testing

Does it continue to work consistently?

Reliability testing evaluates whether a product continues to perform its intended function consistently over time and under expected operating conditions.

PERFORMANCE

How well does it work?



RELIABILITY

Does it continue to work that well?

How Can We Test Reliability?

1

Repeated Operation

Perform the same operation repeatedly.

e.g. 500 pick-and-place cycles

2

Duration

Operate continuously for a defined period.

e.g. Electronics operating 8 hours

3

Changing Conditions

Test across realistic variations.

e.g. Loads, lighting, temperature, battery

4

Failure Observation

Observe what fails, when, and how frequently.

Performance vs Reliability

Domain	Performance	Reliability
Computer Science	How quickly does the application process one request?	Does it keep working correctly after repeated requests / prolonged operation?
Electronics	What is the measurement error?	Does measurement remain consistent during prolonged operation?
Mechanical	Can the mechanism lift the required load?	Can it repeatedly lift the load without unacceptable degradation?
Automation	What is the positioning accuracy?	Does it maintain acceptable accuracy over repeated cycles?

MINDSET SHIFT

Think Like a Failure Investigator

How can your MVP fail? Possible examples:

Sensor failure

Mechanical wear

Loose connections

Overheating

Battery depletion

Software crash

Communication failure

Misalignment

Component fatigue

Performance degradation

Environmental variation

Don't design tests to prove your MVP is good — design them to discover where it fails.

Try to Break Your MVP... On Paper

- Step 1: Identify 3 likely failure modes of your MVP.
- Step 2: Design a reliability test that could expose each failure.
- Step 3: Define test method, operating conditions, cycles/duration, metric, and acceptance criterion.

Possible Failure	Reliability Test	Conditions	Cycles/Duration	Metric	Acceptance Criterion

A reliability test that cannot expose a failure tells you very little.

Engineers Have a Usability Problem

- You give your MVP to a user.
- The user struggles.
- You say: “No, no... click here first.”
- Then: “Select this.”
- Then: “Now press START.”
- The user completes the task.

Did your product pass
the usability test?

NO.

*The engineer just became
part of the product.*

Usability Testing

Can the intended user use it effectively?

Usability testing involves giving representative users realistic tasks and observing how effectively they can use the product.

What can we observe or measure?

Task
completion

Time taken

Errors made

Number of
attempts

Assistance
required

Points of
confusion

User
perception

Watch what representative users actually DO with your MVP — not what they say about it.

A usability test is only useful when the task is realistic and the user is representative.

Who Should Be Your Usability Test Participant?

Choose people who:

- ✓ Belong to the intended user group
- ✓ Were not involved in developing the product
- ✓ Are reasonably unfamiliar with how the MVP works

Your teammates cannot be your usability test participants.

They already know:

- How the product works
- What the controls mean
- What the intended workflow is
- What the expected result should be

They are not fresh users.

Don't Ask. Observe.

WEAK USABILITY TEST

"Is our product easy to use?"

User:

"Yes."

BETTER USABILITY TEST

Give the user a realistic task:

"Use this system to complete ____."

Then observe:

- Did they succeed?
- How long did they take?
- Where did they hesitate?
- What errors did they make?
- Did they require assistance?

What users DO can reveal more than what users SAY.

Usability Across Engineering Domains

Computer Science

TASK

Create an account and complete a key workflow.

OBSERVE

- Completion
- Time
- Errors
- Navigation difficulties

Electronics

TASK

Connect the device and obtain a measurement.

OBSERVE

- Incorrect connections
- Setup time
- Assistance required

Mechanical

TASK

Adjust the mechanism for a different workpiece.

OBSERVE

- Adjustment time
- Errors
- Difficulty handling controls

Automation

TASK

Configure and start an automated operation.

OBSERVE

- Setup errors
- Sequence confusion
- Assistance required

Example of a Usability Test

PRODUCT

Portable water-quality testing device

REPRESENTATIVE USER

Farmer

TASK

"Use this device to test this water sample and determine whether the water is suitable for irrigation."

Metric	Acceptance Criterion
Task completion	≥90% users
Completion time	<3 minutes
Critical errors	0
Assistance required	None
Ease-of-use rating	≥4/5

Put Your MVP in Someone Else’s Hands

- Identify: WHO are your representative users? WHAT 2–3 realistic tasks will they perform? What will you OBSERVE or measure, and what constitutes a PASS?

Representative User	Task	What Will You Observe?	Metric	Acceptance Criterion

Your teammates are NOT usability test participants.

**It Works. It Is Reliable.
It Is Easy to Use.**

But the customer still says...

“NO.”

- X Costs ₹3 lakh when the customer expects ₹1 lakh
- X Occupies twice the available space
- X Requires excessive maintenance
- X Produces unacceptable noise
- X Cannot integrate with the customer's existing system

Technically successful does not automatically mean acceptable.

User Acceptance Testing

Will the intended user/customer accept it?

USABILITY

Can the user use it effectively?

USER ACCEPTANCE

Does the product satisfy the user's/customer's requirements sufficiently for actual adoption/use?

Acceptance may depend on:

Functionality

Performance

Reliability

Ease of use

Cost

Safety

Size/weight

Maintenance

Compatibility/integration

Operating environment

Other app-specific needs

Usability ≠ User Acceptance

Example: Electronics Product

The device is:

- Easy to operate
- Accurate
- Compact

Usability Test: PASS

Can the intended user operate the device correctly?

But...

The battery lasts only 4 hours.

The customer requires operation throughout an 8-hour work shift.

User Acceptance: FAIL

A usable product can still be unacceptable.

User Acceptance Across Engineering Domains

Computer Science

Software works well but cannot integrate with the customer's existing system.

Electronics

Device performs accurately but battery life does not meet field requirements.

Mechanical

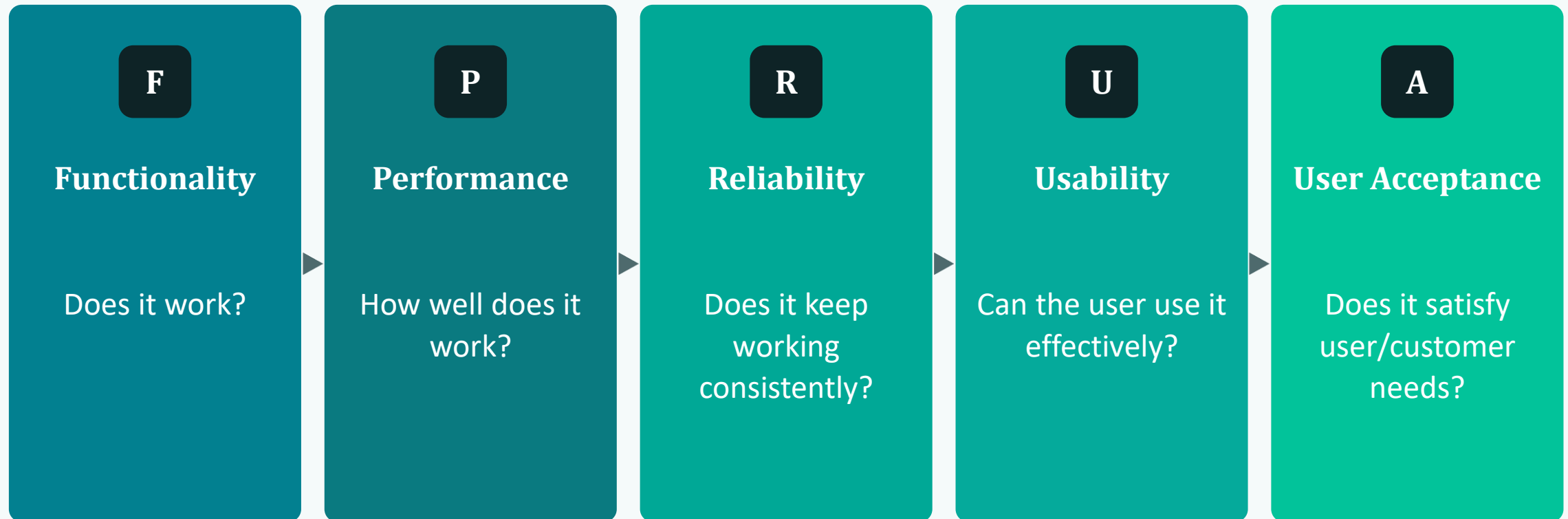
Machine performs the task but its footprint exceeds available workspace.

Automation

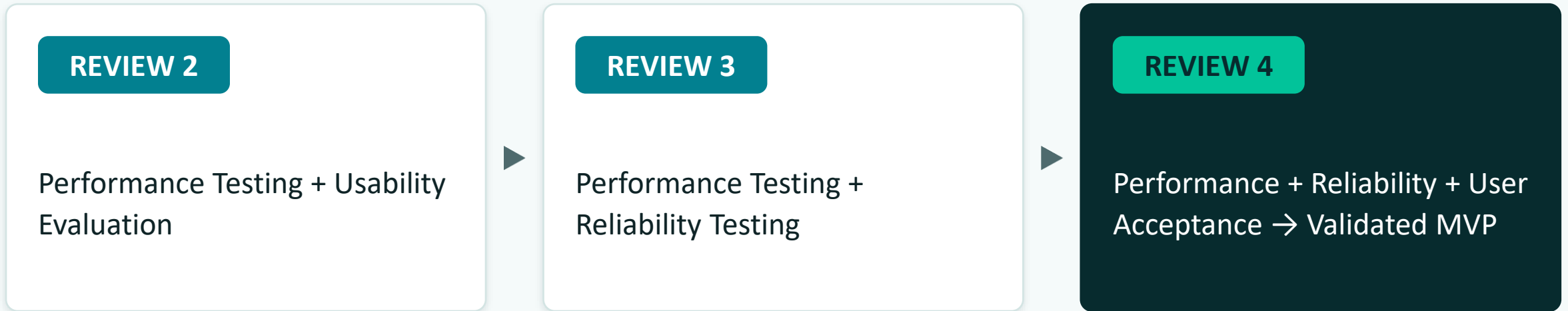
System achieves required throughput but changeover time is unacceptable.

Acceptance is determined in the context of actual use.

From Prototype to Validated MVP



Your E-Co-Op Testing Journey



This progression follows the testing and evaluation sequence specified in the 7th-semester E-Co-Op lesson plan.

Build Your Testing Roadmap

Review all the tests your team developed today. Classify them:



Can Start Now

Tests you can conduct immediately



Need Resources

Equipment, facilities or materials required



Need Users

Representative users must be identified



Need Time

Repeated operation or long-duration testing required



Need Customer / Industry Access

Acceptance testing requires external stakeholders

Identify the THREE tests your team must start first.

CLOSING THOUGHT

Stop Saying...

"Our MVP Works."

Instead, show us:

What did you test?

How did you test it?

Under what conditions?

How many times?

Against what acceptance criterion?

What were the results?

Engineering claims require evidence.